Project Management Plan for THEIA

Revised Project Plan

Version 1.1, Draft 1

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Revision History

Name	Date	Reason for Changes	Version
Initial Draft	08/27/2026	Initial draft from template.	1.0, Draft 1
PPI: Preliminary Plan	09/10/2026	Editing to add preliminary plan content.	1.0, Draft 2
PPI: Revised Plan	09/30/2026	Editing to make revisions suggested.	1.1, Draft 1

1. Overview

1.1. Project Purpose, Objectives, and Success Criteria

The primary purpose of Project THEIA is to build a smartphone application that assists blind and visually impaired people with indoor navigation. Functional objectives focus on getting a user from one location to another inside or across buildings, including detecting obstacles, instructions for turning and displaying distance, and calling emergency services. Non-functional objectives target safe and fast navigation, with an interface that's easy to use and not reliant on readability of the screen. The success for Phase I is determined by successfully completing requirement engineering activities and delivering system specifications and a prototype. Our GitHub repository is available here: https://github.com/muliahulia/CPTS484-Project-Theia.

1.2. Project Deliverables

1. Project Phase I: Preliminary Project Plan (Due September 20)
2. Project Phase I: Final Submission (Due October 18)
3. Project Phase II: Final Submission (Due December 13)
4. Project Demo (Due December 7 - 13)

1.3. Assumptions, Dependencies, and Constraints

The functionality of the system assumes that the user has a smartphone with the necessary sensors and voice recognition abilities. The project is constrained by the need to design around non-functional requirements, like safety and user friendliness, since traditional interfaces aren't usable for the primary stakeholders. A way to create successful documentation is by establishing traceability that's accurate and connecting problems to goals, objectives, and detailed requirement specifications.

1.4. References

- Washington State University, "Cpt S 484-Project THEIA-Preliminary Definition," CPTS 484 Course Material, Pullman, WA, 2026.
- Washington State University, "Cpt S 484 Team Project Specification Phase I," CPTS 484 Course Material, Pullman, WA, 2026.
- "Designing accessible apps for the blind and low vision community," www.letsenvision.com. https://www.letsenvision.com/blog/designing-accessible-apps-for-the-blind-and-low-vision-community (accessed Sept. 16, 2026).
- F. P. Brooks, "No Silver Bullet: Essence and Accidents of Software Engineering," Computer, vol. 20, no. 4, pp. 10-19, Apr. 1987.
- G. Jackson, M. Kosinski, and J. Holdsworth, "What Is the Software Development Lifecycle (SDLC)?," *Ibm.com*, Dec. 13, 2024.

1.5. Definitions and Acronyms

- AS-IS/TO-BE Scenarios: A way to present system features by comparing the current state (AS-IS) to the future state (TO-BE).
- Gen AI: Generative Artificial Intelligence. Tools that may be used to assist the team, requiring disclosure and evaluation by the team.
- GUI: Graphical User Interface. The visual and interactive interface for the app's mockup, which must be carefully designed to support our users.
- NFRs: Non-Functional Requirements. System goals like speed, safety, and comfort.
- THEIA: The name for the indoor navigation smartphone app.
- WRS: World, Requirement, Specification. The main Phase I submission document that clarifies domain and system descriptions while also creatin traceability from problems to specifications.

2. Project Organization

2.1. Process Model

The Project THEIA team will use an agile life cycle model though out the project. An agile model will ensure that the main project requirements will be analyzed and refined multiple times as we learn more about the THEIA system.

Agile Stage	Roles	Activities	Entry Criteria	Exit Criteria
Project Initiation	Team Liaison, All team members	Review project description and preliminary definition documents. Assign roles, establish communication methods, setup repository, and define a project schedule.	Groups are assigned and all necessary documents are received.	Responsibilities are known. Project scope and preliminary plan are established.
Product Development	All team members	Identify requirements and issues iteratively, while following the Problem -> Goal -> Objective -> Requirement specification breakdown. Conduct peer reviews and revisions.	Project Initiation is completed, and responsibilities/roles are assigned.	Phase I documents are drafted and reviewed for final submission
Product Release	Team Liaison, All team members	Final peer review is completed (Professor & Team Review). Deliverables and requirements are verified. Submit Phase I.	Phase I documents are reviewed and completed.	Requirements are verified, and Phase I is submitted.
Project Termination	All team members	Complete required supporting documents. Prepare for the next phase.	Phase I deliverables are submitted	Closing documents are submitted. The team is prepared for the next project phase.

Table F-1: Process Model

2.2. Organizational Structure

The team consists of Vijay Deva, Aryan Puthran, and Julia Smith, who serves as the Team Liaison. The team will use an organized shared drive to collaborate, share project resources, and distribute individual work. General project decisions, process, and work will be reviewed and discussed in the team's selected primary communication, Discord. This will allow members to be held accountable for their deliverables, and major decisions to be reviewed by the entire team.

The Team Liaison will be the main point of communication between the project development team and the main stakeholders. The Liaison will also be responsible for keeping the team on schedule and resolving external/internal issues.

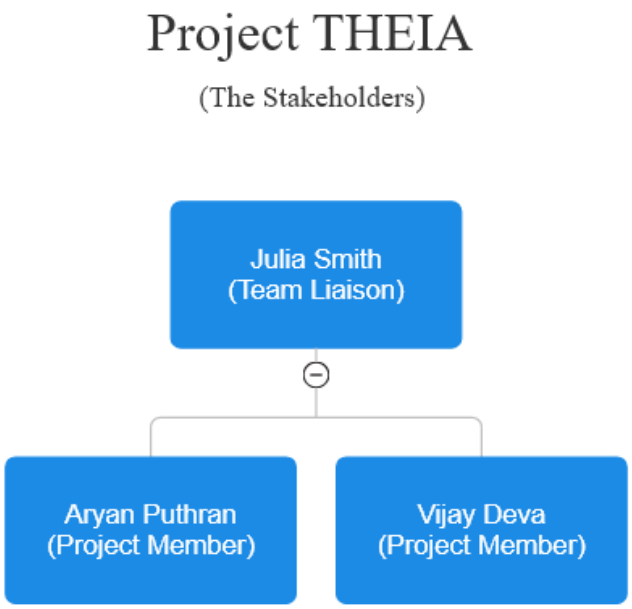


Figure F-1: Organization Chart

2.3. Roles and Responsibilities

Project responsibilities will be distributed equally among all types of group members. Each member will have their individually assigned task as well as their own part in group review sessions. Julia Smith, our Team Liaison, will be responsible for communicating with external stakeholders and keeping the project group on schedule.

Role/Activity	Description	Person/Owner
Team Liaison	Primary communication for team and stakeholders. Help coordinate deliverables based on project deadlines.	Julia Smith
Project Member	Develop/maintain individually assigned requirements and deliverables. Participate in group	Julia Smith, Aryan Puthran, Vijay Deva

	review sessions. Interact and communicate in the team's primary communication platform (Discord).	
AS-IS/TO-BE Scenarios & Presentation	Develop scenarios and a presentation based on the template provided in Canvas. Scenarios should outline current problems with the THEIA application functionality.	Julia Smith, Aryan Puthran, Vijay Deva
WRS Document	Main submission for Phase I. Includes introduction, issues, WRS, prototype, and traceability matrix.	Julia Smith, Aryan Puthran, Vijay Deva
Identify and rework Software Issues/Requirements	Read through informational requirements and identify any issues within the THEIA functionality. Resolve/rework any issues that are found.	Aryan Puthran, Vijay Deva
Document Issues/Requirements	Document how issues found within the THEIA functionality can be resolved and reworked.	Julia Smith
Prototype and User Manual	Develop the prototype and document how users would interact and efficiently use the application.	Aryan Puthran
AI Usage and Teamwork Report	Create a report with evidence (screenshots, chat logs) of how the group utilized GenAI. If AI is not used, explain why.	Vijay Deva

Table F-2: Project Responsibilities

3. Managerial Process Plans

3.1. Management objectives and priorities

The primary objective for Phase I is to transform the informal Preliminary Definition into a comprehensive WRS document, supported by a GUI mock-up prototype, a preliminary user manual, and a recorded presentation.

Priorities in order:

1. Every issue identified in the Preliminary Definition must trace a Problem, which traces a Goal, which decomposes Objectives, which are addressed by Requirement Specifications.
2. Issues must be identified across all three sections of the Preliminary Definition (Domain/Stakeholders, Functional, and Non-Functional).
3. Since work is distributed among team members, the team keeps the final document consistent.
4. Handle the AI Usage Report, if applicable, as work happens, not as an afterthought.

3.2. Assumptions, dependencies, and constraints

Assumptions

- The Preliminary Definition document provided is intentionally informal/incomplete, and the team is expected to fill gaps using reasonable, justified design decisions rather than seek official clarification for every ambiguity.
- Team members have access to a shared git repository and will consistently meet virtually to collaborate.
- No physical hardware, functional sensors, or backend infrastructure will be developed for Phase I. The mockup is a concept-level artifact only.

Dependencies

- The WRS document's "Improved Understanding" section depends on the "Issues" section being completed first.
- The AS-IS/TO-BE presentation depends on the Top Priority TO-BE scenario being identified from the WRS analysis.
- The User Manual depends on the TO-BE scenarios and the prototype mock-up being finalized, since it must walk users through those scenarios using the app as a mocked up.
- Final submission depends on instructor feedback.

Constraints

- All deliverables must follow the provided templates on Canvas.
- The Project Plan may only include deliverables relevant to Phase I.
- The recorded presentation is capped at 10-15 minutes.

- Any use of Gen AI must be fully disclosed, evidenced via screenshots and chat logs, and evaluated.
- Course deadlines are fixed and non-negotiable.

3.3. Risk management

The following risks have been identified based on the project's unique challenges, including team coordination, system requirements, and the usage of AI tools.

Risk Description	Potential Impact	Mitigation Strategy
Inconsistent Interpretation of Definitions	Team members may have different interpretations of the Preliminary Definition, leading to conflicting requirements.	Holding regular team meetings to discuss and ensure we're aligned on interpretations before drafting documents.
Weak Traceability	Failing to connect Problems to Goals, Objectives, and Requirement Specifications will degrade the WRS document.	Implement iterative peer reviews focused on verifying traceability across all documentation.
Uneven Workload or Missed Meetings	Lack of team commitment or busy schedules can result in uncoordinated or late submissions.	Ensure the Team Liaison enforces the schedule, use Discord for accountability, and ensure tasks are distributed fairly.
Trouble Finding Real Users	The inability to interview actual stakeholders could lead to an app that misses critical accessibility needs.	If real users cannot be found, the team must use "creative imagination" to step into the shoes of a blind user to determine usability and safety needs.
Overreliance on Gen AI	While we don't plan on it, blindly submitting AI output without refining or checking it can lead to errors or a negative impact on grades.	We don't plan to use Gen AI, but should we use it, we will treat it as an assistant. All Gen AI outputs will be reviewed, refined, evaluated, and reported in the AI Usage report (if applicable).

Table F-3: Management of Risks

4. Technical Process Plans

4.1. Methods, tools, and techniques

Methods

- Requirements Engineering: Utilizing the AS-IS/TO-BE scenario technique to evaluate current navigational challenges and outline desired system improvements for the THEIA app.
- Modeling Traceability: We'll follow the Problem → Goal → Objective → Requirement Specification, breakdown to build the WRS document, with traceability maintained between each level.
- Agile Process: Implementing an iterative Agile life cycle to analyze, draft, and refine project requirements as understanding deepens.

Tools

- Version Control: A shared GitHub repository will be used to host and keep copies of all project documents.
- Collaboration & Communication: A shared drive will be used for working together and file sharing. Discord will serve as the main platform for team communication.
- Presentation: PowerPoint will be used to construct the AS-IS/TO-BE slides using the provided template, and YouTube will be used to host the recorded presentation.
- Prototyping: GUI mock-up software will be used to make a concept level interface tailored for our stakeholders.

Techniques

- Team members will use imagination to put themselves in the position of a potential blind user, ensuring the requirements and mock-ups address the needs of visually impaired individuals.
- Sections will undergo peer review before being integrated into the final document to catch inconsistencies early and unify the team's vision.
- The team will have frequent virtual meetings to evaluate progress and resolve disagreements regarding interpretation of the requirements.

4.2. Software documentation

This project will primarily focus on using the provided WRS template as the central piece of documentation. The WRS document will capture detailed system requirements, ensuring we use traceability from problems to specific functional and non-functional requirements. To better support this, the team will also make AS-IS/TO-BE scenario slides to outline system features, and a user manual that guides users through the TO-BE interactions using the GUI prototype.

Because this phase is to exercise software requirement engineering, the team will avoid creating documentation for later, like testing plans or detailed architecture design documents.